



DesignWare® Cores

SuperSpeed USB 3.0 Controller

Release Notes

*DWC_usb3 – **Product Codes***

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Release Notes

These release notes describe the changes for Version 3.30b of the DesignWare® USB 3.0 Controller core.

The following topics are covered:

- [“Product Codes”](#) on page 4
- [“Changes from the Last Release”](#) on page 5
- [“Fixed Defects and Enhancements \(STARs on the Web\)”](#) on page 6
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1.1 Product Codes

Table 1-1 lists the products and product codes for the DWC_usb3 controller.

Table 1-1 Product Codes for DWC SuperSpeed USB 3.0 Controller

Component Name or Add-On	Product Code
Base Products	
DWC USB 3.0 Device-AHB ¹	6793-0
DWC USB 3.0 Host-AHB ²	6897-0
DWC USB 3.0 Dual Role Device-AHB ³	7567-0
DWC USB 3.0 Hub	6921-0
DWC AP USB 3.0 Device	C234-0
DWC AP USB 3.0 Host	C233-0
DWC AP USB 3.0 Dual Role Device	C232-0
Add-Ons	
DWC USB 3.0 Device-Isoch Add-On	7571-0
DWC USB 3.0 Host-Isoch Add-On	7593-0
DWC USB 3.0 DRD-Isoch Add-On	7568-0
DWC USB 3.0 Hibernation Add-On	9228-0
DWC USB 3.0 HSIC Add-On	7387-0

1. DWC USB 3.0 Device-AHB includes DWC USB 3.0 Device-AXI Add-On and DWC USB 3.0 Device-Isoch Add-On.

2. DWC USB 3.0 Host-AHB includes DWC USB 3.0 Host-AXI Add-On and DWC USB 3.0 Host-Isoch Add-On.

3. DWC USB 3.0 Dual Role Device-AHB includes DWC USB 3.0 DRD-AXI Add-On and DWC USB 3.0 DRD-Isoch Add-On.

1.2 Changes from the Last Release

Only packaging-related updates are made in version 3.30b of the DWC_usb3 controller. These updates are to fix the PVE simulation failures for certain configurations (see STAR 9001350747).



Note

There are no RTL updates in this release.

Changes Made in 3.30a Version of Controller

The following updates were made in version 3.30a of the controller (for your reference):

- RTL
 - Enhancements listed in [Table 1-2](#)
 - Bug fixes (for details, see “[Fixed Defects and Enhancements \(STARs on the Web\)](#)” on page 6)
- The coreConsultant simulation environment now supports PVE (instead of VTB)
- SSIC and OTG features are deemphasized (no longer supported)

Table 1-2 Enhancements

STAR#	Brief Description
Device Enhancements	
9001273158	ADC: Ref_clk based SOF
Host Enhancements	
9001249644	ENH: Link ECN: U1 Residency Timer
9001227980	Host: USBCMD[HCRST] De-assertion Behavior Change
9000980770	Faster Recovery to SS.INACT Transition for Port Configuration Timeout
9001216358	Host: Delay Disabling DBC when LSP is Busy
9001206584	Enhancement to Improve USB 2.0 Host IPD when FAST_TAT is Disabled
9001205997	ADC 3.0 Changes
9001207904	Optimizing USB2 PHY Suspend Behavior in Host Mode
9001252670	Option to Increase FS/LS/SUSPEND Disconnect Time from 2.5us to 32us
9001285211	Programmability for Detecting LFPS Signaling Duration for U1/U2 Exit
9001250761	Change PORTSC.PLS along with PORTSC.PED
9001296792	Enhancement to Avoid Host Not Responding with Shorted Cable

Table 1-2 Enhancements (Continued)

STAR#	Brief Description
DRD Enhancements	
9001242267	Support New SS PHY Power States (P3.CPM/P4)
9001247356	Support ref_clk Gating
9001247358	Disable RX Detection Enhancement
9001131280	Synchronizer Replacement with BCMS

You must replace all the source files with the new files to get these changes. For more details on the changes, refer to the following:

- Source file changes: Use src/DWC_usb3.list for list of source files
- Programming model changes: See the change-bar version of the Databook
- Register changes: See the change-bar version of the Databook
- Port name changes: See the change-bar version of the Databook

1.3 Fixed Defects and Enhancements (STARs on the Web)

You can view a complete list of problem reports for this product, including problems identified after product release, by accessing the STAR report on the Web. You must have a SolvNet ID to view STAR reports. For a list of STARs for the USB 3.0 Controller, go to the following links:

- Device: https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_device
- Host: https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_host
- Hub: https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_hub
- DRD: https://www.synopsys.com/dw/ipdir.php?c=dwc_usb_3_0_drd

For Automotive Package:

- https://www.synopsys.com/dw/ipdir.php?c=dwc_ap_usb_3_0_device
- https://www.synopsys.com/dw/ipdir.php?c=dwc_ap_usb_3_0_host
- https://www.synopsys.com/dw/ipdir.php?c=dwc_ap_usb_3_0_drd

You can access STAR reports for any DesignWare verification model or synthesizable component through the IP directory:

<http://www.synopsys.com/designware/ipdir>

Enhancement STARs are identified by the letter “E” in the Severity column. Fixed defect STARs are identified by the letter “B.” For previous core versions, click the tab for the version in question.

1.4 Quality Metrics and Inter-Operability Testing

1.4.1 Hardware Testing

The following hardware and software testing is performed on the USB 3.0 Controller.

**Note**

Hardware validation uses the Synopsys HAPS51FXT FPGA platform.

1.4.1.1 Device

Hardware Testing for DWC_usb3 Device Features

- Control endpoint
- Bulk IN
- Bulk OUT
- ISOC OUT Audio (speaker) - With USB 2.0 Host and USB 3.0 Renesas Host
- ISOC IN Audio (microphone) - With USB 2.0 Host and USB 3.0 Renesas Host
- Interrupt - Synopsys CBI application
- Synopsys BOT application
- Synopsys UASP application with MCCI xHCI driver - Preliminary

Hardware Testing for Interoperability

- Compliance Tests
 - The Device controller passed USB 3.0 Device compliance
 - The Host passed USB 3.0 xHCI Host compliance
- Hosts Tested
 - Synopsys xHCI Host
 - TI xHCI Host
 - Renesas xHCI Host
 - Fresco xHCI PDK Host
 - AsMedia xHCI Host
 - USB2.0 High Speed Hosts
 - USB2.0 Full Speed Hosts
 - Intel USB 3.0 Motherboard
 - AMD USB3.0 Motherboard
 - VIA
 - Etron

- PHYs Tested
 - Synopsys USB 3.0 PHY (PIPE and UTMI)
 - TI USB 3.0 PHY (PIPE and UTMI and ULPI)
 - Xilinx RocketIO configured as USB 3.0 SS PHY
 - Synopsys USB2.0 UTMI PHY
- Device Drivers
 - Synopsys Mass-Storage BOT and UASP Linux driver
 - Synopsys ISOC Speaker and Microphone Linux driver
 - MCCI device stack
- Host Driver Tested
 - Win8 driver
 - Open source xHCI Driver (BOT)
 - PDK's Windows Driver from USB-IF
 - Renesas Windows driver
 - Fresco Windows driver
 - MCCI Driver
 - USB 2.0 Windows EHCI driver

1.4.1.2 Host

- Devices tested
 - USB 3.0 devices: Synopsys, TI, Realtek, Buffalo, Western-Digital, Fujitsu, SIIG, LucidPort, Eagle Consus, Bytecc, Sharkoon, Kingston, Seagate, SuperTalent, AData, Kingston, VIA, Genesys, Blackmagic, JMicron, Evolution.
 - 250+ USB2.0 Devices covering HS/FS Thumb-drives, HS/FS HDDs, HS/FS Cameras, Printers, Scanners, Multifunction Printers, MP3 Players, Cellphones, Photo-Albums, HS/FS Webcams, Wired/Wireless Keyboards, Mouse, Headsets, Microphones, Biometric Scanners, HS/FS Hubs, WiFi/Ethernet adapters, Wired/Wireless Game-controllers, DVD Players, Broadband Modem, Barcode Scanner, Display Adapters, Sound Cards, TV Tuners, and so on.
 - Device interoperability tested in Linux and Windows
- Hubs tested
 - Synopsys
 - TI
 - VIA
 - SIIG
 - Genesys
 - Evolution

- Host drivers
 - Opensource Linux driver
 - MCCI xHCI driver with Microsoft BOT class - Windows
 - MCCI xHCI driver with MCCI BOT class - Windows
 - MCCI xHCI driver with MCCI UAS class - Windows
 - USB-IF xHCI Driver with Microsoft BOT - Windows
 - USB-IF Compliance Test Driver
- xHCI-Debug Feature: Host interoperability tested
 - Intel Chipset xHCI Host
 - Synopsys xHCI Host
 - Renesas xHCI Host
 - Fresco xHCI Host
 - VIA xHCI Host
 - AsMedia xHCI Host

1.4.1.3 Hub

Tested with:

- Hosts tested
 - Synopsys xHCI Host
 - TI xHCI Host
 - Renesas PDK Host
 - Fresco Host
- Devices tested
 - USB 3.0 Devices: Synopsys, TI, Buffalo, Western-Digital, SIIG, LucidPort, Eagle Consus, Bytecc, Sharkoon, Kingston, SuperTalent, and AData
- Host Driver Tested
 - MCCI Windows
 - Renesas Windows Driver
 - MCCI Driver
 - Linux Driver
 - PDK's Windows Driver from USB-IF

1.4.1.4 DRD

DRD is hardware tested by loading the Device and Host drivers at a given time. There is no combined DRD driver.

1.4.1.5 LPM Errata Testing

LPM Errata is tested using Synopsys Host and Synopsys Device with Microsoft Win8 xHCI driver.

1.4.1.6 HSIC Testing

HSIC is tested with Synopsys HSIC PHY. Interoperability is tested between DWC_usb3 host and DWC_usb3 device, and also between DWC_usb3 host/device and DWC_hsiotg device/host.

1.5 Known Issues and Limitations

This section describes the known issues and limitations.

1.5.1 Feature and Verification Limitations

- Device
 - In device mode, the Device does not support a bInterval setting of 15 or 16 (which equates to a polling interval of 2^{14} or 2^{15} microframes). The ITP frame counter is 14 bits wide, so in general USB devices cannot detect service interval boundaries unless they can count more than 0x3FFF microframes. Host driver software (for example, the USB D) typically polls devices every 32 microframes regardless of whether they request a slower polling interval. In addition, it is not clear that there is an application that requires a polling interval greater than 2^{14} microframes.
 - There is no limitation in using a Device in Mass-Storage application for BOT and UASP class, Periodic (ISOC and Interrupt), and ethernet class applications.
 - Only control endpoint-0 is supported. Support for multiple control endpoints is not needed for most device applications. Though support for multiple control endpoints is coded, this configuration has not completed the entire control abort error testing.
- Host
 - The host controller does not check for ISOC buffer overrun, bandwidth error, and parameter error conditions. The software driver should not create these programming error conditions (which is the expected functional operation).
 - The host controller does not generate Endpoint Not Enabled Error when an invalid doorbell is rung. The RTL ignores all invalid doorbell rings.
 - The host controller does not adjust the bus interval automatically based on the received Bus Interval Adjustment Notifications from the devices. A workaround for this limitation is described in the section "Host Mode – Miscellaneous Topics" in Chapter 3, "Integrating the Core", of the *DesignWare Cores SuperSpeed USB 3.0 User Guide*.
 - The host controller does not support multiple Event Data TRBs in a single Transfer Descriptor (TD). In a TD, the Event Data TRB must be the last TRB with the Chain bit set to '0'. In addition, the host controller has limited support for Event Data TD's (an Event Data TRB that is not chained at the end of a TD; the TD consists of just one Event Data TRB). In some scenarios, the EDTLA reported from an Event Data TD is incorrect.
 - The host controller does not calculate the system bus bandwidth requirement when it handles Configure Endpoint commands. Software must calculate the system bus bandwidth requirement of all the endpoints that it intends to configure. It must also ensure that the system bus bandwidth requirement does not exceed the bandwidth that the system can provide.

- ❑ The xHCI Specification requires that software setup the entire TD for an Isochronous IN endpoint before Isochronous Scheduling Threshold (IST). However, it does not require that software always setup $\text{MaxPacketSize} * (\text{MBS}+1) * (\text{Mult}+1)$ bytes of TD. The controller requires both.
- ❑ xHCI Extended Capabilities:
 - I/O Virtualization is not supported.
 - Local Memory is not supported.
 - Debug Capability can be enabled only when the Master Bus Data Width is greater than 32 bits.
- ❑ xHCI Debug Capability: There is a known issue involving the setup bytes in a control transfer when the Master Bus Data Width (DWC_USB3_MDWIDTH) is 32 bits, therefore, the xHCI Debug Capability must not be selected when MDWIDTH is 32.
- ❑ P3 in U2 (GUCTL1[25]) is not supported if SOFITPSYNC and LPM_SYNC are 0.
- ❑ With 3.00a release of the controller, the xHCI Errata, Contiguous Frame ID capability (CFC) is added. With this, a legacy xHCI driver not updated for CFC and setting up ISOC TDs with SIA=0 could result in some Missed Service Error from the xHC. This is a known behavior. With CFC enabled, xHC cannot distinguish between legacy drivers setting SIA=0 and a CFC compliant driver setting SIA=0.
- ❑ When using 16-bit UTMI and multiple ports, the host controller does not meet the USB turnaround time with five Hubs.
- Hub
 - ❑ The core is a SuperSpeed-only Hub product. You need a companion USB 2.0 Hub.
 - ❑ JTAG mode is not hardware or software tested.
- Unsupported features:
 - ❑ IC-USB, 3-pin/6-pin USB1.1 serial interface, I2C, and CarKit functions are not supported.
 - ❑ AXI xm_awcache, xm_arcache, xs_awcache, and xs_arcache are not supported.
 - ❑ Synchronous reset mode is not supported.
 - ❑ SSIC
 - ❑ OTG
 - ❑ ADP
- Usage of the GUCTL[13] bit in simulation with Synopsys VIP
 - ❑ Simulations that use Synopsys USB VIP fail during low-power exit if the GUCTL[13] bit is set to '1' because the VIP does not currently support this feature. Therefore, the GUCTL[13] bit is set to '0' in the PVE tests to override the GUCTL parameter settings.
 - ❑ There is no functional issue when the GUCTL[13] bit is set to '1'. This functionality is validated in hardware.
- Synthesis

In host-only mode, when "Enable UTMI-16bit Fast Turnaround Operation" is enabled (DWC_USB3_FAST_TAT_EN = 1), the mac_clk is synchronous to the utmi_clk/ulpi_clk. There should not be any set_false_path -hold and set_max_delay between the mac_clk and the

utmi_clk/ulpi_clk. However, the sdc files generated by the coreConsultant still have them set between the mac_clk and the utmi_clk/ulpi_clk; these need to be removed. This hold issue affects release 2.60a and earlier.

- Host Simulation and Hardware Validation Summary ([Table 1-3](#))

Table 1-3 Host Features – Simulation Coverage Status

Feature	Simulation Coverage
Non Periodic Transfers (Bulk and Control)	Yes
Stream (Primary and Secondary)	Yes
Multi-Port (2-port, and 4-port)	Yes
Multiple Bus-Instance - SS	Yes Recommended only if your SoC bandwidth is larger than total USB 3.0 bandwidth to sustain maximum USB 3.0 transfer rate (Per port: 4Gbs IN + 4Gbs OUT + 480Mbps + 12Mbps). Also SoC “latency + access time” for a 1KB packet is less than the USB 3.0 transfer time (Latency < 2.1uS/Number of SS Bus Instances). Not meeting these conditions could result in not meeting Periodic transfer quality of service (QOS).
Multiple Bus-Instance - HS	Yes
Connect/Disconnect	Yes
Suspend/Resume	Yes
LPM	Yes
HW-LPM	Yes
U1/U2/U3	Yes
Non Periodic Split	Yes
SS Hub	Yes
HS Hub	Yes
FS Hub	Yes
Periodic Transfers (ISOC and Interrupt)	Yes
Periodic Split	Yes
Periodic Ping	Yes
Extended Features	
■ Supported Protocol Capability	Yes
■ xHCI Message Interrupt Capability (MSI)	Yes
■ Secondary Streams	Yes
■ Legacy SMI Support	Yes
■ Number of Ports override (host_num_u2/u3_ports)	Yes
■ Port Disable (host_u2/u3_port_disable)	Yes

Table 1-4 Host Features – Hardware/Software Interop Testing Status

Feature	Hardware Interop Testing
Non Periodic Transfers (Bulk and Control)	Yes
UASP	Yes
Multi-Port (2-port)	Yes
Multi-Port (4-port)	Yes
Multiple Bus-Instance - HS	Yes (2 HS Bus-Instances)
Multiple Bus-Instance - SS	Yes (2 SS Bus-Instances)
Connect/Disconnect	Yes
Suspend/Resume	Yes
LPM	Yes
HW-LPM	Yes
Get Port Bandwidth command for a Secondary Bandwidth Domain	No driver support
U1/U2/U3	Yes
PC Hibernation and Standby	Yes
Non Periodic Split	Yes
SS Hub	Yes
HS Hub	Yes
FS Hub	Yes
Suspend, Resume, and Hibernation	Yes
Periodic Transfers (ISOC and Interrupt)	
■ SS ISOC Device (Synopsys Speaker and Microphone application, BlackMagic)	Yes
■ HS Webcam	Yes
■ FS Webcam	Yes
■ LS Mice	Yes
Periodic Split	Yes
Periodic Ping	Basic PING Tested With Synopsys ISOC Device
Extended Features	
- Supported Protocol Capability	Yes
- xHCI Message Interrupt Capability (MSI)	Yes
- Secondary Streams	No - No driver support
- Legacy SMI Support	(One customer tested BIOS boot supported)
- Number of Ports override (host_num_u2/u3_ports)	Yes
- Port Disable (host_u2/u3_port_disable)	No - No driver support

1.5.2 Known Simulation Environment Problems

- Gate-level simulation may not pass in all configurations.
- When PHY model is used for simulations in coreConsultant, sometimes the packaged testcase may not pass for few configurations. Report this to [Customer Support](#), and continue testing without the PHY in direct connect mode.
- Depending on your configuration, the simulation with PHY GTECH model can run into infinite loop. If you have access to Synopsys PHY RTL, use the PHY RTL for simulation or use USB 3.0 VIP PIPE3 interface.

Known Third-Party Simulator Support Issues

- When you use VCS, the below warning appears many times in the log file:
Warning- [SV-NFIVC] Non-void Function Used In Void Context
There is a known VIP issue to fix these constructors, but there is no functional issue related to this and all tests pass with the warning.
- Hub simulations do not support SuperSpeed GTECH-PHY or Synopsys RTL PHY. Use only VIP PIPE3 interface.

1.5.3 Known Formality Problems

- FM 2010.12-SP3 fails due to SVF format mismatch.
- FM 2010.03-SP3 with DC 2010.12-SP3 may also fail due to SVF format mismatch. Switch to the latest DC. At the time of this release, the latest DC tested is 2010.03-SP3.
- 2010.03 and earlier versions hang in the verification stage in host configurations.
- When you choose "DCTCL_opto_strategy" as the synthesis strategy, formality does not pass for all configurations. Synopsys is aware of this issue and is considering it as a tool interoperability issue. Validate formality using the "Design Compiler Reference Methodology" to confirm the RTL to netlist formal equivalency.
- A few configurations have failures on "logic_analyzer" and "ram_*" ports. This can be a tool issue.

1.5.4 Known Synthesis Problems

- The core dump and also the netlist created by DC 2013.03 is sometimes not readable by TetraMAX tool.
- It is recommended to use the "Design_Compiler_Reference_Methodology" for coreConsultant synthesis. The ATPG coverage with "DCTCL_opto_strategy" is low.
- In some host configurations, the following are warnings on some unused/un-driven wires. The RTL looks clean:

```
Warning: In design 'DWC_usb3', net 'xm_wstrb[0]' has multiple drivers (unknown wired-logic type). (LINT-38)
```
- Error in Formality 2009.06 in coreKit synthesis flow. Although formality passes, the following error message is displayed:

```
Could not find /global/apps3/fm_2009.06/amd64/bin/snpsperl. Check Synopsys installation
```

To avoid this error, use 2008.09-SP4 version.
- Although the scan coverage is 99.7%, there are a few scan violations in this release. The scan violations are listed below:
 - 131 trailing edge port captured data affected by new capture violations (C6), 131 Data path affected by clock captured by clock in trailing edge clock_port violations (D14) (PRE-DFT), 743 Clock path affected by clock captured by clock in level sensitive clock_port violations (D15) (PRE-DFT).
These are due to falling-edge flops in the synchronizers. This does not reduce the coverage.
 - Two clocks connected to primary output violations (C17), 1 clock connected to primary output affected by new capture violation (C18).
The ram_clk_gated clock is an output clock, which is needed for the RAMs. This violation does not appear in a SoC DFT where the ram_clk_gated is not an primary output.
- The coreConsultant GUI currently does not support synthesis run targeting an FPGA using Synplify.

1.5.5 Known coreConsultant Problems

- When you close a workspace in coreConsultant and then create a new workspace during the existing coreConsultant session, the following error occurs:

```
Error: Attribute 'RegisterSize' is not valid on item 'GUSB2PHYCFG_REGS'. (CMDS-140)
```

Workaround: When you close a workspace and you want to create a new one, you must exit and restart coreConsultant.
- Constraint Analysis activity under Verify Component is not supported.

1.6 Web Resources

- DesignWare IP product information: <http://www.designware.com>
- Your custom DesignWare IP page: <http://www.mydesignware.com>
- Documentation through SolvNet: <http://solvnet.com> (Synopsys password required)
- Synopsys Common Licensing (SCL): <http://www.synopsys.com/keys>

1.7 Customer Support

To obtain support for your product, choose one of the following:

- Enter a call through SolvNet.
 - Go to http://solvnet.synopsys.com/support/open_case.action, and provide the requested information, including:
 - Product: DesignWare Cores
 - Sub Product: USB 3.0 Device, USB 3.0 Host, USB 3.0 Hub, or USB 3.0 DRD
 - Version: 3.30b
 - Subject: USB 3.0
 - Problem Type:
 - Priority:
 - Title: <insert DWC_usb3>
 - Description: For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood
- Send an e-mail message to support_center@synopsys.com.
 - Include the Product name, Sub Product name, and Version (product release number) in your e-mail so it can be routed correctly.
 - Provide the following additional information, if applicable:
 - For environment setup problems, run the `debug_info` command in the coreConsultant GUI and include the text file generated.
 - For configuration failures, include the error messages that appear in the coreConsultant GUI console pane.
 - For simulation failures, include a text file with your specific configuration. Generate this text file using the coreConsultant GUI's `write_batch_script` command. Also include the log file from the <workspace>/sim/SoC_sim/pve directory, the log file for the specific test, and a VPD/VCD waveform dump file.
 - For synthesis failures, include the log file from the <workspace>/syn/ directory.
- Telephone your local support center.
 - North America:
Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
 - All other countries:
<https://www.synopsys.com/support/global-support-centers.html>